

IA: Investigating the Effect of Salinity of Dissolved Oxygen

Research Question

To what extent does the salinity (0, 10, 20, 30, 40, 50 g/dm³) of water affect the dissolved oxygen (ppm) using the Winker Titration Method

Candidate Number:

Word Count: 2795

Introduction

Climate change and global warming have become the one of the most pressing issues of the 21st century altering ecosystems around the world. Taking Environmental Systems and Societies (ESS) in IB, I became interested in aquatic ecosystems, and I began to read journals and books on how global warming is affecting marine organisms. Due to the changes in inflows of freshwater, which directly affect the salinity of the water, ichthyoplankton species, fishlike salmon or tilapia, and invertebrates are changing their migration habits, and which has severely impacted aquatic food chains¹. Not only does this affect our current consumption of fish, but because the species that are affected most by salinity are at lower trophic levels, their changes in population affect species with higher trophic levels like birds or larger aquatic organisms².

As I read further, I began to understand that the dissolved oxygen in the water plays a vital role in the key metabolic process (e.g., respiration) of many marine organisms, and connecting these 2 phenomena, I was interested to explore how saline conditions affect the dissolved oxygen levels: an area of research which hasn't been extensively studied. In numerous aquatic environments such as the Black Sea, dissolved oxygen plummeted with the high salinity, however the incremental changes in salinity affecting dissolved oxygen weren't uncovered³.

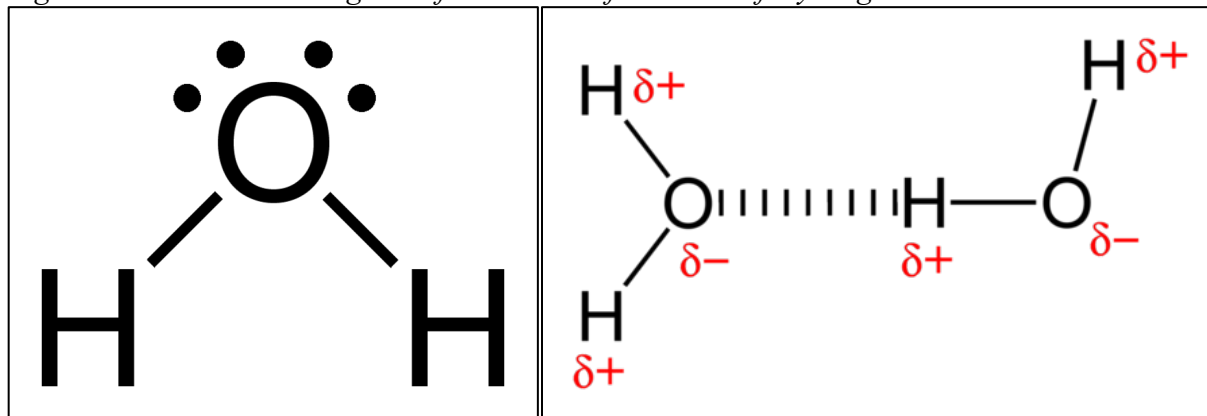
¹ Mukherjee, P., Mitra, A., Zaman, S., & Mitra, A. (2021). Impact of Climate-Change-Induced salinity alteration on ichthyoplankton diversity of Indian sundarbans. In Springer eBooks (pp. 2933–2959). https://doi.org/10.1007/978-3-030-57281-5_276

² Jia, Y., Jiang, Y., Liu, Y., Sui, X., Feng, X., Zhu, R., & Chen, Y. (2021). Understanding trophic structure variation in fish assemblages of subtropical shallow lakes: Combined effects of ecosystem size, productivity, and disturbance. *Ecological Indicators*, 129, 107924. <https://doi.org/10.1016/j.ecolind.2021.107924>

³ Pakhomova, S., Vinogradova, E., Yakushev, E., Zatsepin, A., Shtereva, G., Chasovnikov, V., & Podymov, O. (2013). Interannual variability of the Black Sea Proper oxygen and nutrients regime: The role of climatic and anthropogenic forcing. *Estuarine Coastal and Shelf Science*, 140, 134–145. <https://doi.org/10.1016/j.ecss.2013.10.006>

Background Information

Figure 1 and 2: Lewis Diagram of Water⁴ and formation of Hydrogen Bonds⁵



Water (H_2O) is a covalent molecule with a bent molecular geometry due to the two pairs of non-bonding electrons on the oxygen. This unequal sharing of electrons (resulting from the unequal electronegativity of hydrogen and oxygen) causes the hydrogen atoms to remain slightly positive (δ^+) and the oxygen retaining a slightly negative charge (δ^-), which creates a dipole moment (see figure 1). This separation of charge allows water molecules to form hydrogen bonds— a type of intermolecular force— where the slightly positive hydrogen is attracted to the lone pair causing a strong force of attraction (see figure 2). A water molecule can form up to 4 hydrogen bonds with neighboring molecules and thus the sheer number of hydrogen bonds form the basis of water's high melting point and water solvent properties, being coined the 'universal solvent'⁶.

Non-polar gases like oxygen dissolve via weak van de Waals forces, and thus their solubility remains limited. Specifically, dipole-induced dipole forces form when oxygen dissolves in water. The highly polar water molecules induce a temporary dipole on the non-polar oxygen, polarizing the electron cloud and creating a weak intermolecular force. Therefore, dissolved oxygen is the amount of free oxygen gas not bound to other elements present in water⁷. Numerous bodies of water in the world consist of salt, and I wanted to understand the effects of salinity on dissolved oxygen.

Thus, my research question for this investigation was:

To what extent does the salinity (0, 10, 20, 30, 40, 50 g/dm³) of water affect the dissolved oxygen (ppm) using the Winker Titration Method

⁴ TheChemGuys. (n.d.). Lewis Diagrams. <http://thechemguys.blogspot.com/2011/06/lewis-diagrams.html>

⁵ CK-12 Foundation. (n.d.). CK-12 Foundation. <https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/section/9.19/primary/lesson/hydrogen-bonding-chem/>

⁶ genchem. (n.d.). https://www.westfield.ma.edu/cmase/gen_chem2/imf/hbonds/hbond.htm#:~:text=Hydrogen%20bonds%20are%20a%20special,N%2C%20O%2C%20or%20F.

⁷ Dissolved oxygen | US EPA. (2024, October 24). US EPA. [https://www.epa.gov/caddis/dissolved-oxygen#:~:text=Dissolved%20oxygen%20\(DO\)%20refers%20to,by%20aquatic%20plants%20during%20photosynthesis.](https://www.epa.gov/caddis/dissolved-oxygen#:~:text=Dissolved%20oxygen%20(DO)%20refers%20to,by%20aquatic%20plants%20during%20photosynthesis.)

Hypothesis and Null Hypothesis

This study hypothesized that as salinity increased, the dissolved oxygen concentration would decrease. When salt in the form of NaCl dissolves in water, they form intermolecular bonds which compete with oxygen to interact with and thus reduce the overall interactions between oxygen and water molecules, which reduces the dissolved oxygen concentrations.

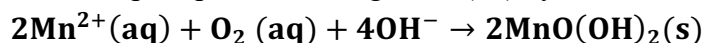
In this case, the null hypothesis would state that there will no relationship between the salinity of water and concentration of dissolved oxygen.

Methodology

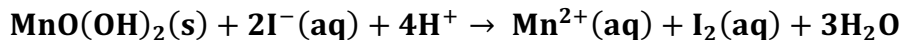
Sequence of Reactions

The Winkler Titration method was used to determine the dissolved oxygen (DO) present in water through a sequence of redox titrations. The following sequence of reactions occur, to determine the dissolved oxygen:⁸

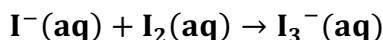
1. First the Manganese(II) sulfate is added to water in a basic medium to fix the dissolved oxygen and forms a brown precipitate of Manganese (III) hydroxide:



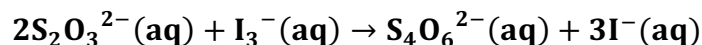
2. Sulfuric acid is added which acidifies the solution and Mn (III) hydroxide oxidizes iodide ions to iodine



3. Iodine equilibrium is reached for tri-iodine ions which enhance the titration accuracy



4. The iodine produced is titrated with sodium thiosulfate, and the color change is yellow to colorless



⁸ King, W. (2011, February 11). Winkler Titrations – Measuring Dissolved Oxygen. Colby at Sea. Retrieved May 28, 2025, from [https://web.colby.edu/colbyatsea/2011/02/11/winkler-titrations-measuring-dissolved-oxygen/#:~:text=Winkler%20Titrations%20-%20Measuring%20Dissolved%20Oxygen%20*,%20S4O6%2D\(aq\)%20+%203%20I-%20\(aq\)%20\(4\)](https://web.colby.edu/colbyatsea/2011/02/11/winkler-titrations-measuring-dissolved-oxygen/#:~:text=Winkler%20Titrations%20-%20Measuring%20Dissolved%20Oxygen%20*,%20S4O6%2D(aq)%20+%203%20I-%20(aq)%20(4))

Variables

<i>Independent Variable</i>	<i>Units</i>	<i>Range</i>	<i>Method to Manipulate</i>
Salinity of water	g/dm ³	0, 10, 20, 30, 40, 50	Exact mass balance was used to measure mass of NaCl
		The concentration range was selected to include intervals of 10 g/dm ³ which allows accurate identification of the relationship between salinity and DO. Also, the specific concentrations were based on the salinities of real bodies of water. ⁹	
			Uncertainty of balance = ±0.05 g/dm ³

<i>Dependent Variable</i>	<i>Units</i>
Volume of Sodium Thiosulphate needed to decolorize solution which was used to derive the concentration of Dissolved Oxygen (DO)	Volume of Sodium Thiosulphate was measured in decimeters
	DO measured in parts per million
	5 trials were conducted for each solution to reduce the impact of random error and increase the reliability of the results

⁹ Webb, P. (n.d.). 5.3 salinity patterns. Pressbooks. <https://rwu.pressbooks.pub/webboceanography/chapter/5-3-salinity-patterns>

<i>Control Variables</i>	<i>How it was controlled</i>	<i>Impact on Reaction</i>
Temperature of water	All water samples were taken at room temperature at 25°C	An increase in the temperature would decrease the dissolved oxygen and vice versa respectively and prevent accurate results from being obtained, outlined by Henry's Law ¹⁰ .
Volume of water used	250cm ³ of water was measured by a measuring cylinder	Any deviations in the volume of water would either increase or decrease the concentration of solute and thus change the DO.
Volume and concentration of Winkler reagents	2.3M of MnSO ₄	All concentrations of all reactants were carefully weighted using a balance and solution were prepared using volumetric flask. If there were any deviations in any of the reagents, the specific amount of moles of oxygen wouldn't be able to be accurately obtained.
	0.6M of Alkaline KI	
	0.01M of Na ₂ S ₂ O ₃	
	1% of Starch solution	
Time between adding reagents	10 minutes were waited before titrating the solution	Due to reaction kinetics, time is required for the reaction to complete, and thus a stopwatch was used to ensure 10 minutes passed before the reaction continued
Light exposure during reaction	Bottles were kept in a shaded area during the reaction	Because iodine can photodegrade with excess light, bottles were kept in a dark room. ¹¹
pH and purity of water	Tap water was being used from the same source throughout the experiment	Changes in the tap water consistency could affect DO

¹⁰ Pomelo. (2024, July 2). How does temperature affect dissolved oxygen? | Atlas Scientific. Atlas Scientific. https://atlas-scientific.com/blog/how-does-temperature-affect-dissolved-oxygen/?srsltid=AfmBOop0NOevAojU-Ntf6O6MPcFSkzoawl6Jap4K_slcOB-G2SUaQ31e

¹¹ Figueiredo, A., Strekowski, R., Bosland, L., Durand, A., & Wortham, H. (2020). Photolytic degradation of molecular iodine adsorbed on model SiO₂ particles. The Science of the Total Environment, 723, 137951. <https://doi.org/10.1016/j.scitotenv.2020.137951>

Apparatus

Mass balance	± 0.05 g
100 mL measuring cylinder	± 0.5 mL
250 mL measuring cylinder	± 1.0 mL
250 mL beaker	
Stirring rod	
thermometer	$\pm 0.5^{\circ}\text{C}$
Ice bath	
Dropper	
1 mL graduated pipette	± 0.05 mL
250 mL stoppered reagent bottles	
250 mL conical flask	
Burette stand + clamp	
White tile	

Method Used

The method is taken from CLEAPSS:¹²

Preparing Solutions

1. Standard method for making volumetric solutions
 - a. Weight required amount of solute using weighing balance
 - b. Dissolve solid in small volume of distilled water in a beaker and stir until no solid remains
 - c. Then transfer to a volumetric flask using a funnel
 - i. Rinse the beaker, rod and funnel with small portions of water and add the rinses to the flask
 - d. Make up to the mark with water so meniscus sits on the calibration line at eye level
 - e. Add stopper and invert several times
2. Solution A
 - a. Use method to create standard solutions using 39g of Manganese (II) sulphate and 180cm³ of distilled water
3. Solution B
 - a. Use method to create standard solutions using 8g of sodium hydroxide pellets and 70cm³ of distilled water
 - b. Once stock solution is created, cool the solution to below 20°C in an ice bath and then add another 8g of sodium hydroxide (measured from a mass balance)
 - c. Repeat until a total of 32g of sodium hydroxide has been added
4. Sodium Thiosulfate solution

¹² RB105 - Winkler's method for dissolved oxygen. (n.d.). <https://science.cleapss.org.uk/resource-info/rb105-winkler-s-method-for-dissolved-oxygen.aspx>

- a. Use method to create standard solutions using 4.96g of sodium thiosulphate pentahydrate and 2 liters of distilled water
5. Starch solution
 - a. Use method to create standard solutions using 1g of sodium thiosulphate pentahydrate and 100mL of distilled water
6. Different saline solutions
 - a. 0, 10, 20, 30, 40, 50 g/dm³ of NaCl were prepared
 - i. 1.6 dm³ of water was used to create these solutions and the mass of sodium needed was determined using table 1:

Concentration \g/dm ³	Mass of Sodium \g
0	0
10	16
20	32
30	48
40	64
50	80

Measuring DO

1. A 250cm³ sample of water from each concentration was collected and added to a stoppered bottle
 - a. Ensure no air is trapped inside
2. Use a 1cm³ pipette to transfer 1cm³ of Solution A to the water sample
 - a. Inject Solution A below the surface of the water
 - b. Careful not to introduce air to the water
3. Use a similar technique from step 2 to transfer 1cm³ of Solution B to the water sample
4. Replace the stopper on the bottle without trapping air inside
5. Agitate the bottle to mix the reagents and then allow the bottle to stand for 10 minutes
 - a. Brown manganese (III) hydroxide is expected to precipitate
6. Transfer the contents of the bottle to a larger flask
7. Add 1.5cm³ of concentrated sulfuric acid using the pipette
 - a. The precipitate will dissolve, and the iodine will be liberated
8. Titrate the solution with standardized sodium thiosulfate in the burette
 - a. Mark the initial and final volumes of sodium thiosulphate
9. Add 2cm³ of the starch solution near the endpoint:



Sample after adding starch



End point of titration

10. Repeat steps 1-9 for each salinity of water
11. Repeat steps 1-10 to obtain 5 trials for each salinity

Safety, ethics and environmental issues

<i>Substance</i>	<i>Concern</i>	<i>Solution</i>
Concentrated sulfuric acid	Safety issue – can severely burn skin and eyes and can cause respiratory problems ¹³ Ethical issue – corrosive chemical	Wear gloves and eye-protection and avoid direct contact. Perform in fume hood. Store sulfuric acid and sodium hydroxide in corrosive cabinet
Manganese sulphate	Safety – both are irritant to skin and eyes Environmental –can disrupt ecosystems and change aquatic oxygen demand	Handle with gloves, eye-protection and lab coat
Potassium iodide		
Sodium thiosulphate		
Starch solution	Ethical – Use minimal amounts as used in food	Prepare small, fresh batches to avoid waste
Sodium chloride	Safety – irritant to eyes Ethical – use minimal amount as used in food Environmental – excess salt discharge can cause salinization in ecosystems	Use only necessary and dispose in dilute solutions down sink with excess water
Sodium hydroxide	Environmental issue – improper disposal could contaminate biological ecosystems Safety – highlight corrosive and can cause chemical burns ¹⁴	Dilute the sodium hydroxide with lots of water and dispose down the drain. Wear gloves and eye protection when handling to avoid infection.
Broken Glass	Safety issue – can cause cuts to skin Environmental issue – broken glass can contribute to lab waste	Use gloves and eye-protection and use broken-glass bins to minimize waste
Water usage	Ethical issue – lots of water is consumed during the experiment	Minimize waste and reuse water where appropriate

Data Presentation

Sample Calculation

The mass required to prepare the solutions of Alkaline KI, MnSO_4 , $\text{Na}_2\text{S}_2\text{O}_3$, and the starch was calculated using the following equations:

¹³ Sulfuric acid DCCEEW. (2022, August 15). [https://www.dceew.gov.au/environment/protection/npi/substances/factsheets/sulfuric-acid#:~:text=Health%20effects-,Description,the%20lungs%20\(pulmonary%20oedema\).](https://www.dceew.gov.au/environment/protection/npi/substances/factsheets/sulfuric-acid#:~:text=Health%20effects-,Description,the%20lungs%20(pulmonary%20oedema).)

¹⁴ Sodium hydroxide | Medical Management Guidelines | Toxic Substance Portal | ATSDR. (n.d.). <https://wwwn.cdc.gov/TSP/MMG/MMGDetails.aspx?mmgid=246&toxid=45>

$$\text{Concentration} = \frac{\text{Moles}}{\text{Volume}}$$

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar Mass}}$$

For example: Preparing 0.6 M Alkaline KI. 0.25dm³

Using the concentration formula, the moles required is calculated:

$$0.6 \times 0.25 = \text{moles}$$

Thus 0.15 moles are required, and the mass required is calculated given that the molar mass of 166.01 g/mol.

$$0.15 \times 166.01 = \text{mass}$$

Thus, the mass required is 24.09 grams

Table 1: Raw data for volume of Na₂S₂O₃ needed for different salinities

Concentration of sample \ g/dm ³ (±0.05)	Volume of Na ₂ S ₂ O ₃ used as titer \ cm ³ (±0.05)						Moles of Na ₂ S ₂ O ₃ (mM)
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average volume	
0.00	2.30	2.50	2.40	2.10	2.40	2.34	0.23
10.00	2.30	2.20	1.90	2.40	2.40	2.24	0.22
20.00	1.90	1.80	1.90	2.30	2.10	2.00	0.20
30.00	1.80	1.90	2.10	2.00	1.80	1.92	0.19
40.00	1.30	1.70	1.90	1.70	2.10	1.74	0.17
50.00	1.60	1.60	1.70	1.90	1.50	1.66	0.17
60.00	1.40	2.10	1.50	2.00	1.50	1.70	0.17
70.00	1.20	1.30	1.50	1.70	1.40	1.42	0.14

1. Volume needed to neutralize was calculated by subtracting the final reading of the burette from the initial reading
2. All 5 trial readings for each concentration were analyzed using the ± 2 standard deviations (z-scoring method) and it was determined that there were no statistical anomalies
3. The Average volume of each trial was computed using the following formula where x_i represents each individual value and n represents the number of values:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

a. E.g., $\frac{2.30+2.50+2.40+2.10+2.40}{5} = 2.34\text{cm}^3$

4. Moles of Na₂S₂O₃ calculated using the following formula, Given the concentration of Na₂S₂O₃ was 0.01M. The volumes were given in cm³ and converted to dm³

$$\text{Concentration} \times \text{Volume} = \text{Moles}$$

a. E.g., $0.01 \times 0.00234 = 0.000234 \text{ moles}$ which equals 0.23 mM

The overall total uncertainty was calculated by adding the individual percentage uncertainties. To calculate the total absolute uncertainty, the following formula was used:

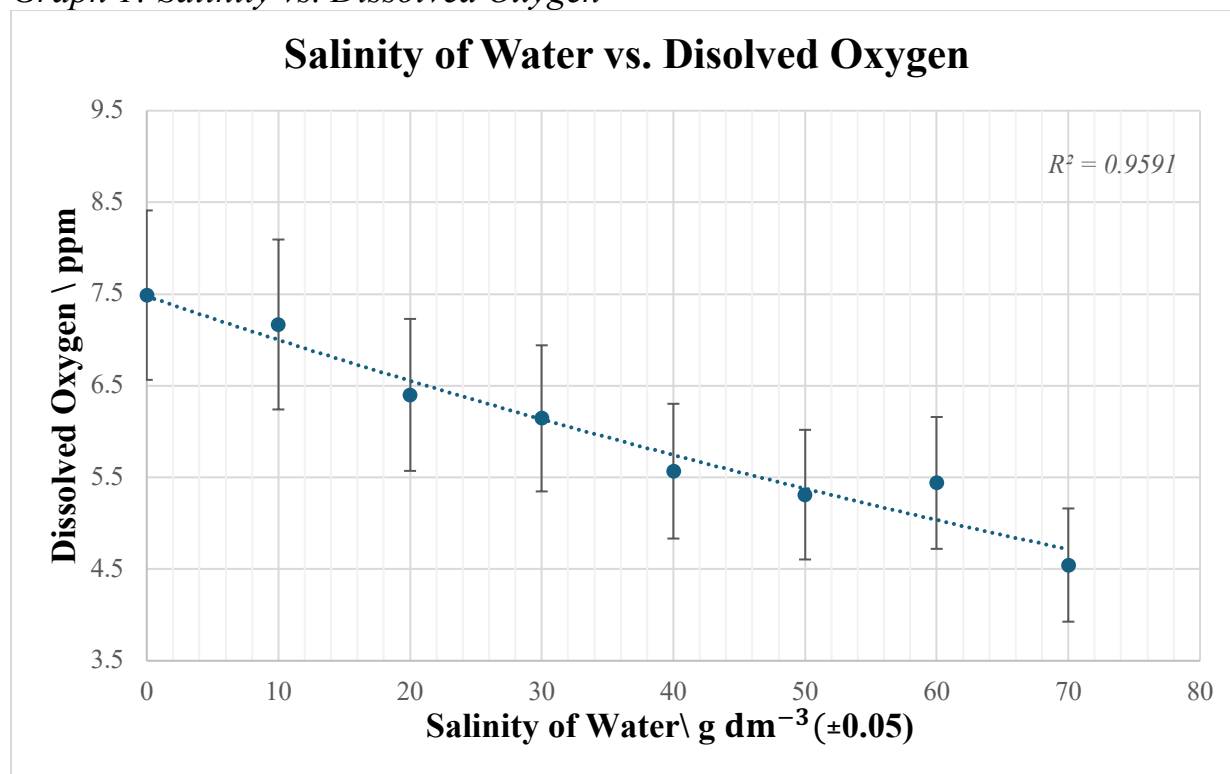
$$\frac{\% \text{ Uncertainty}}{100} \times DO$$

Thus, for example, when the salinity was 0 g/dm^3 , the absolute uncertainty was $\frac{12.34}{100} \times 7.48 = 0.924$

Table 4: Absolute Uncertainty Calculations

Salinity \ g/dm^3 (± 0.05)	Dissolved oxygen concentration \ ppm
0.00	7.49 ± 0.92
10.00	7.17 ± 0.93
20.00	6.4 ± 0.83
30.00	6.14 ± 0.80
40.00	5.57 ± 0.74
50.00	5.31 ± 0.71
60.00	5.44 ± 0.72
70.00	4.54 ± 0.62

Graph 1: Salinity vs. Dissolved Oxygen



*Error bars represent absolute uncertainty calculated above

Table 6: Outlining calculation of sum of least square residuals

Concentration of sample g/dm ³ (±0.05)	Actual concentration of dissolved oxygen ppm	Predicted concentration by trendline	Residual	Residual Squared
0.0	7.488	7.475	-0.013	0.000
10.0	7.168	6.969	-0.199	0.039
20.0	6.400	6.498	0.098	0.010
30.0	6.144	6.059	-0.085	0.007
40.0	5.568	5.649	0.081	0.007
50.0	5.312	5.267	-0.045	0.002
60.0	5.440	4.911	-0.529	0.280
70.0	4.544	4.579	0.035	0.001
Sum of Least Square Residuals = 0.346				

The sum of squared residuals was calculated using the formula $\sum_{i=1}^n (y_i - f(x_i))^2$, where Σ = sum of, n is the upper limit, y_i is the actual concentration of dissolved oxygen and $f(x_i)$ is the predicted value by the trendline. The closer this value to 0, the better the fit of the trendline.

The R^2 coefficient was calculated by the equation $1 - \frac{RSS}{TSS}$ where RSS was the sum of squares of residuals and TSS was total sum of squares. The R^2 coefficient was 0.959.

Conclusion

The research question states:

To what extent does the salinity (0, 10, 20, 30, 40, 50 g/dm³) of water affect the dissolved oxygen (ppm) using the Winker Titration Method

The overall trend outlined by the raw and processed data highlights that the increase in salinity of water reduces the dissolved oxygen concentration. Table 1 highlights how decreased volumes of sodium thiosulphate were needed when the concentration of salt in water increases, which is supported by the processed data in table 2 and the graph. Initially, 7.49 ppm of dissolved oxygen was present in pure tap water, and the dissolved oxygen concentration exponentially decays to 6.14 ppm in 30 g/dm³ and eventually reaches 4.54 ppm at the highest salinity assessed being 70 g/dm³.

Salts like NaCl disassociate into Na⁺ and Cl⁻ ions. These ions are highly polar and strongly interact with the water molecules through dipole-dipole forces compared to the weaker intermolecular force between oxygen and water molecules being dipole induced dipole forces. Because these dipole– dipole interactions are stronger, they out-compete the oxygen gas to

interact with water and thus reduce its solubility in water. Also, when water solvates salt ions, it forms hydration shells which reduce the availability of free water molecules for oxygen to interact with. Finally, as NaCl begins to dissolve in the water, it reduces the mole fraction of water which lowers the chemical potential and vapor pressure of water. This aligns Raoult's Law and results in a decreased capacity of the water to dissolve nonpolar gases¹⁵.

Also, the R^2 coefficient being 0.959 being very close to 1 suggests the strong fit of the exponential trendline modeled between salinity of water and its dissolved oxygen. This coupled with the sum of least square residuals being 0.363 highlights that the exponential decay model used to fit the trendline was an accurate model of this relationship.

Evaluation

Overall, the percentage uncertainties ranged from 12.3% to 13.6% which is acceptable compared to the literature review conducted. This uncertainty reflects the use of standard volumetric equipment and the possibility of methodological error.

According to the literature review, the theoretical concentration of dissolved oxygen when no salt is present roughly at 25°C is 8.26 ppm¹⁶. In contrast, this study obtained that there existed 7.49 ppm of dissolved oxygen in pure tap water. The difference between the theoretical and actual values highlights the unreliability and possibility for error in this study. Using this, the percentage error was calculated using this formula:

$$\% \text{ Error} = \frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} \times 100$$

$$\text{Hence } \frac{8.26 - 7.49}{8.26} \times 100 = 9.32\%$$

Therefore, considering that the percentage error is less than the percentage uncertainties, this means that the results fall within the margin of error expected based on the equipment and the method. Thus, this highlights that there weren't any significant systematic errors and thus random error either instrumental or procedure accounts for the discrepancy. Some possible instrumental errors that influenced the results could be the pipettes used. The small volume errors that could have occurred when adding the Alkaline KI, MnSO_4 , or the starch solution, could have easily shifted the stoichiometry and caused random error. Also, air bubbles being present in the sample bottle could have artificially increased the DO by adding extra oxygen which could have deviated the results and caused procedural error. Therefore, these results highlight the real impacts that salinity has on biological ecosystems. Whether it be aquatic organisms like fish or plankton and coral, the increasing salinity of the bodies of water across the globe is changing migration and directly affecting ecosystems.

¹⁵ Wikipedia contributors. (2025a, March 18). Raoult's law. Wikipedia. [https://en.wikipedia.org/wiki/Raoult%27s_law#:~:text=Raoult's%20law%20\(%25CB%88r%25C9%91%25CB%90u%25CB%90lz%2520law\)%20is%20a%20relation,by%20its%20mole%20fraction%20in%20the%20mixture.](https://en.wikipedia.org/wiki/Raoult%27s_law#:~:text=Raoult's%20law%20(%25CB%88r%25C9%91%25CB%90u%25CB%90lz%2520law)%20is%20a%20relation,by%20its%20mole%20fraction%20in%20the%20mixture.)

¹⁶ Hach Company, Goldman, C. R., U.S. E.P.A., & Wetzel, R. G. (n.d.). Dissolved oxygen (DO) [Report]. https://www.gov.nt.ca/sites/ecc/files/dissolved_oxygen.pdf

Strengths	Source of error	Solution
The trend/pattern in the results are clear as the sum of least squares were low, with a very strong positive R^2 .	Indeed there were some variations between trials, but none were statistical anomalies. Some of the variations between trials might have arose due to the subjective endpoint detection of the Winkler titration. This might have introduced some error when determining the precise volumes of sodium thiosulphate	Use a colorimeter to improve the consistency and standardize the assessment of when the endpoint was reached.
7 independent variable values were used which provides enough datapoints to establish a clear trend.	Oxygen might have entered the sample or air bubbles may have been trapped which weren't visible, thus artificially increasing the DO.	Keep the BOD flasks closed and carefully tap the flasks to get rid of air bubbles present.
The apparatus used such as the burette, pipette and the balance had low uncertainties and were sufficiently precise to ensure the research question could be accurately validated.	The room temperature wasn't consistently monitored. Thus, there could have been slight deviations in the temperature of the water samples which would have directly affected the dissolved oxygen.	Conduct the experiment at controlled conditions (e.g., 25°C) either by using a water bath or constantly monitoring the room's temperature
There were minimal systematic errors that affected the results	Considering that iodine decomposes with light, the light intensity during the experiment wasn't thoroughly monitored. This could have significantly affected the volume of sodium thiosulphate needed	Perform all steps using brown bottles to store the solution and in low-light environment within the lab or fume hood.
	Possibly there could have been other ions which could have influenced the reactions taking place in the Winkler method that were present in the tap water.	Use distilled water so the presence of other ions is recued and then oxygen could be manually introduced

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